CSCE 4301 · SPRING 2026 · TEAM G18

FINAL DEMO · MAY 13, 2026

Green*Wave*

Helping ambulances move faster through Cairo traffic.

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Milestone 4 / Final Presentation

Ambulances are stuck in traffic.

CAIRO RESPONSE TIME

8–15 min

Average time for an ambulance to reach the patient.

CARDIAC ARREST SURVIVAL

–10% /min

Per extra minute, survival drops by roughly 10%. This isn't a traffic problem — it's a clinical one.

PREEMPTION SYSTEMS IN EGYPT

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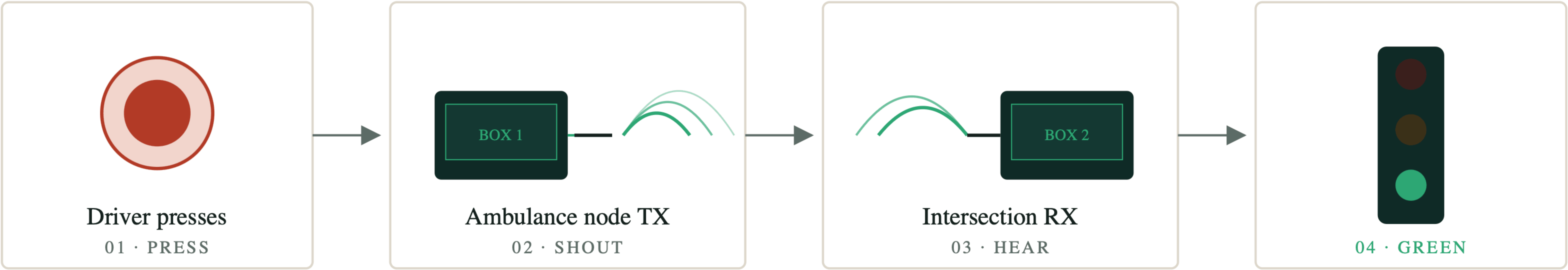
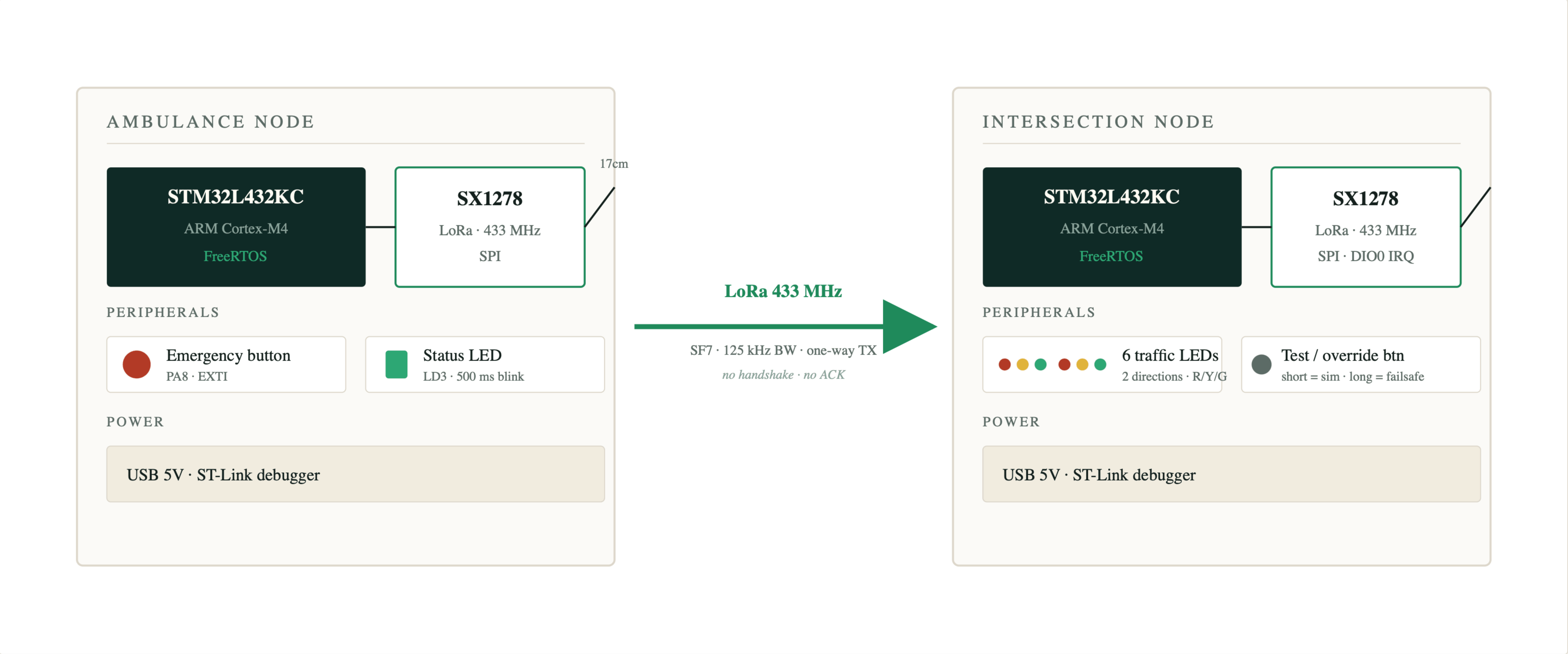
The US, UK, Germany, and Dubai have light-preemption infrastructure. We don't.

Two boxes. One radio link. Lights turn green when an ambulance is coming.



That's it. Release the button and the light goes back to its normal cycle.

Two identical brains, one radio link.



Same brain, different peripherals.

The whole prototype is about **$30** of parts.

AMBULANCE NODE

STM32L432KC Nucleo	brain
Emergency push button	PA8
Status LED	LD3
SX1278 LoRa module	SPI
17 cm wire antenna	¼ λ

ROLE · transmit-only

INTERSECTION NODE

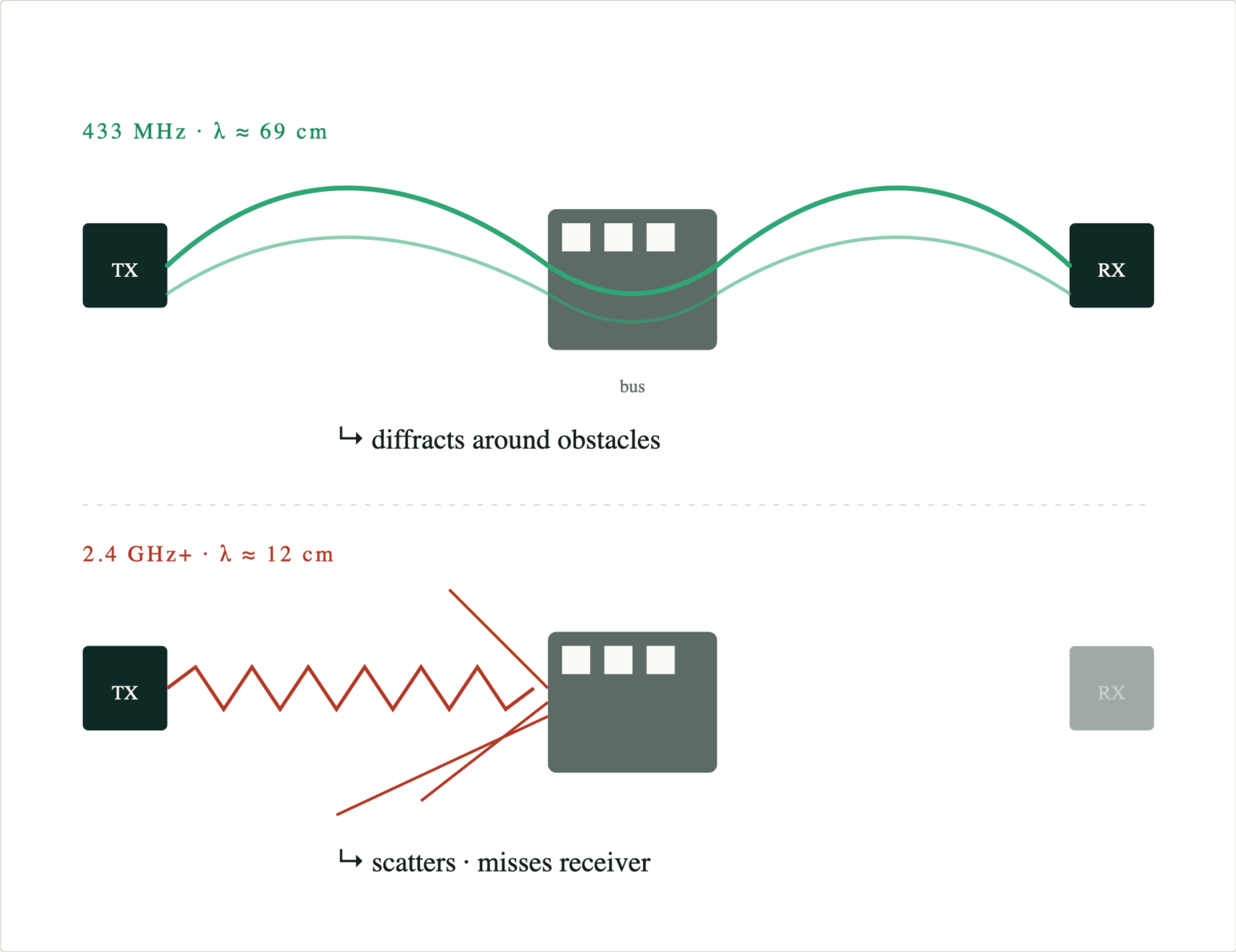
STM32L432KC Nucleo	brain
6 traffic LEDs (R/Y/G × 2)	220 Ω
Test / override button	EXTI
RX indicator LED	PA1
SX1278 LoRa module + antenna	SPI

ROLE · receive + drive lights

Why LoRa. Why 433 MHz.

- **Long range, license-free.** LoRa modulation covers ~1 km in urban clutter.
- **433 MHz** — the open ISM band in Egypt. The proposal's 868 MHz isn't free here.
- **Long wavelength (~69 cm)** physically bends around buses, walls, and bodies. Higher frequencies bounce off everything.
- **SF7 · 125 kHz BW** — a balance of range and packet rate.

> *"That's why we called it GreenWave — the green literally spreads down the road."*



Each packet is 24 bytes — signed and sequenced.

MAGIC	AMB ID	DIRECTION	FLAGS	SEQ #	TIMESTAMP	HMAC-SHA256 (TRUNCATED)
0x47573138	u16	EW	0b01	u32 ↑	u32	tamper-proof signature
4 B · "GW18"	2 B	1 B	1 B	4 B	4 B	8 B

HMAC-SHA256

Nobody can forge a fake "go green."

Receiver re-computes the signature with the shared key — mismatch ⇒ drop.

SEQUENCE NUMBER

Nobody can record a packet and replay it later.

Counter only climbs; receiver rejects anything not strictly greater than the last seen.

Each node juggles four jobs at once.

Interrupts signal tasks via binary semaphores — ISRs stay tiny, work happens in tasks.

Ambulance node

buttonTask	high prio · semaphore
radioTask	every 2 s
sirenTask	500 ms blink
defaultTask	1 Hz heartbeat

← Button EXTI → semaphore → buttonTask

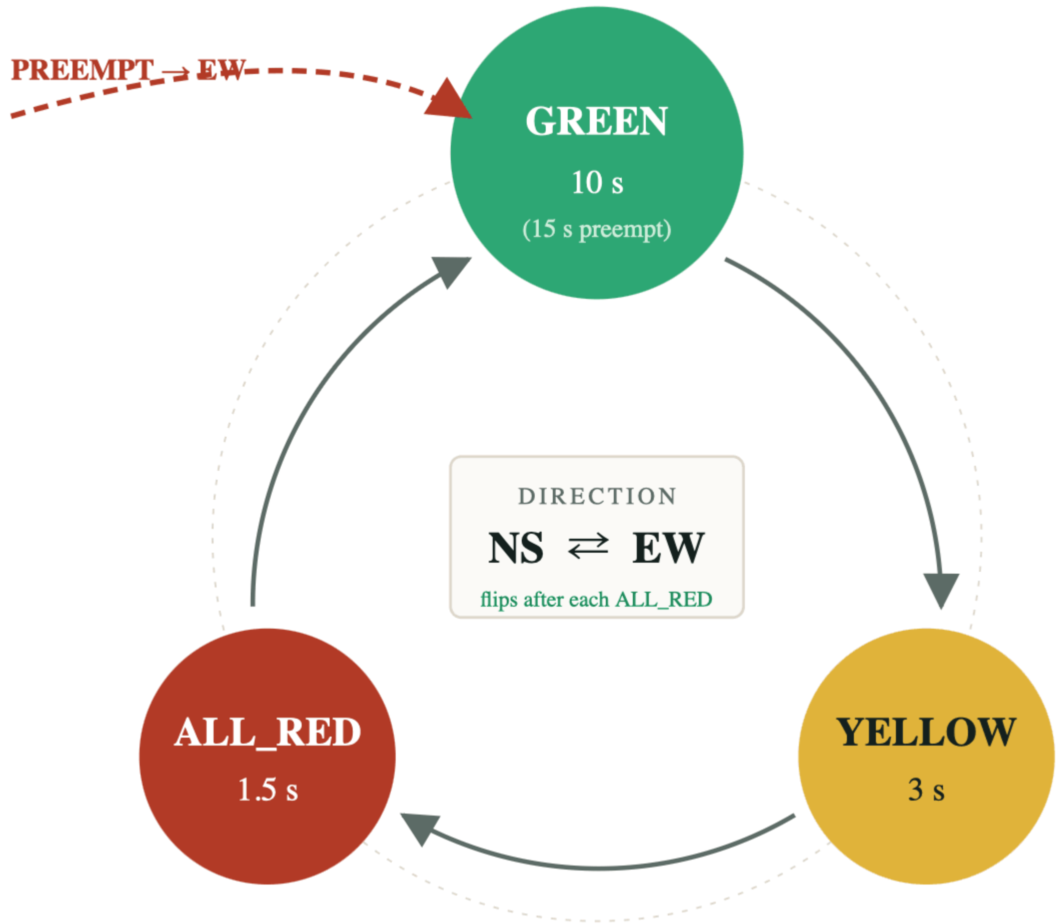
Intersection node

radioRxTask	DIO0 IRQ · semaphore
stateMachineTask	10 Hz
buttonTask	short = preempt · long = failsafe
defaultTask	1 Hz heartbeat

← LoRa DIO0 EXTI → semaphore → radioRxTask

Three phases, mirrored across both directions.

The numbers shown reflect the **preemption cycle** — not the standard cycle. EW green extends to 15 s while an ambulance is signalling.



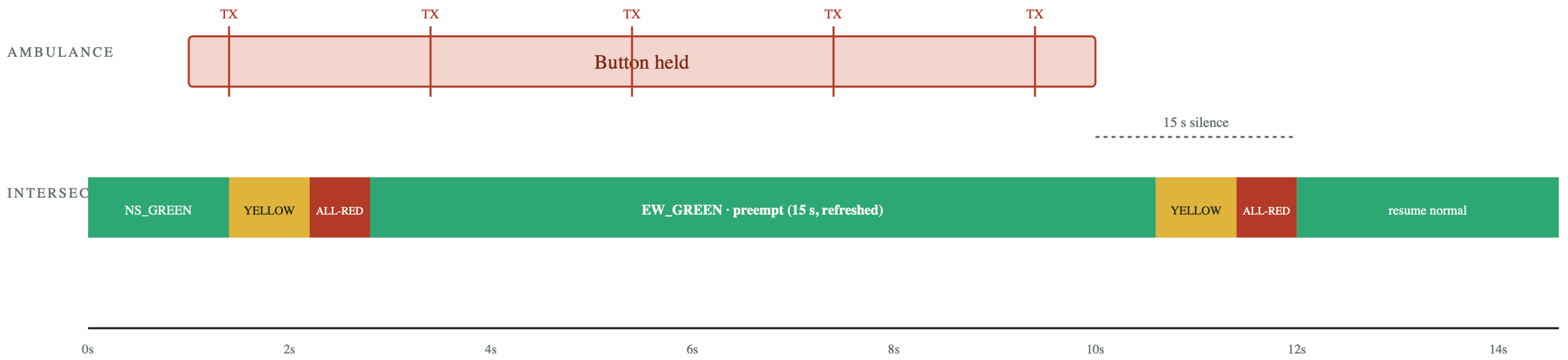
Never green → green.
Every transition passes through yellow + all-red clearance.

● Green ● Yellow ● All-red clearance

CYCLE LEDGER	
GREEN (per direction)	10 s
YELLOW (per direction)	3 s
ALL_RED clearance	1.5 s
EW GREEN during preempt	15 s

> *During preemption, EW green extends from 10 s to 15 s — and refreshes with each new packet.*

The ambulance never causes an unsafe transition.



DURING GREEN

Cut to yellow → all-red → EW green. Never skip clearance.

DURING CLEARING

Let yellow/all-red finish, then snap to EW green.

RELEASE

15 s of silence (~7 missed packets) → resume normal cycle.

When in doubt, fail to a four-way stop.

- **Failsafe** = both reds flash at 1 Hz — drivers treat it as a 4-way stop.
- **Triggers:** LoRa init failure, or operator override.
- **While active:** all preemption requests are blocked.
- **Short-press** the test button = simulated preempt (LoRa fallback).
  Long-press ≥ 2 s = toggle failsafe.

We scoped honestly as we hit real constraints.

PROPOSAL	AS-BUILT	WHY
ESP32 × 2	STM32L432KC × 2	*Course requirement.*
868 MHz	433 MHz	*Egyptian ISM band — 868 MHz isn't license-free here.*
4 approaches · 12 LEDs	2 directions · 6 LEDs	*Tabletop demo scope.*
GPS geofence + directional filter	Hardcoded EW priority	*GPS removed at Milestone 3.*
OLED displays	UART serial logging	*Scope cut at Checkpoint B.*
Tuned soldered antenna	17 cm wire (¼ λ)	*No soldering iron available.*
2nd intersection via I^2C	Single intersection	*Time scope.*

Bench-tested every module, then tested them integrated.

BENCH TESTS (PER SUBSYSTEM)

- Button debounce — clean single events
- TX rate — exactly one packet / 2 s
- Phase timing — within ±50 ms of spec
- LED correctness per phase
- Failsafe activation

INTEGRATION TESTS (WHOLE SYSTEM)

- Preempt during *every* phase transitions safely
- Manual override toggles failsafe on/off
- Multiple consecutive preempts refresh the timer
- LoRa init failure → automatic failsafe

SERIAL · 115200 BAUD

```
[0.012] LoRa init OK · SF7 BW=125kHz
[0.014] FreeRTOS > 4 tasks running
[2.103] RX OK seq=1 rssi=-42 dBm
[2.108] PREEMPT → NS_YELLOW
[5.108] ALL_RED
[6.608] EW_GREEN (hold 15s)
[4.107] RX OK seq=2 rssi=-44 dBm
[6.110] RX OK seq=3 rssi=-41 dBm
[8.114] RX OK seq=4 rssi=-43 dBm
[20.6] timeout → resume normal
[23.1] btn long-press → FAILSAFE
```

> *Diagnostic counters (`dio0_irq_count`, `read_failures`) let us isolate post-RTOS bugs fast.*

Honest scorecard — 6 of 11 fully implemented; 5 partial or revised.

FULLY IMPLEMENTED · 6

- ✓ FR1 FreeRTOS multitasking
- ✓ FR2 Emergency-gated transmission
- ✓ FR5 Safe phase transitions (clearance enforced)
- ✓ FR6 Timeout-based release
- ✓ FR9 Failsafe mode
- ✓ FR10 Manual override

PARTIAL / REVISED · 5

- ◐ FR3 GPS geofence — *removed* (descoped)
- ◐ FR4 Directional filter — *hardcoded* to EW
- ◐ FR7 HMAC — signing works; verify bypassed (bug)
- ◐ FR8 Replay protection — code present, unverified
- ◐ FR11 Display — serial logging, no OLED

14 · NEXT UP

Live Demo

Everything you're about to see is running on the real hardware on the table.

01 Press the ambulance button → LD3 blinks, TX seq 1, 2, 3…
02 Within ~2 s, intersection logs RX OK → cuts to yellow → all-red → EW green.
03 Green is held for 15 s, refreshed while held.
04 Release → TX stops → 15 s silence → resume normal cycle.
05 Long-press test button → failsafe: both reds flash.

The migration was the hard part.

- **Sim → hardware.** At M3 we demoed a Wokwi ESP32 simulation; we then re-implemented the LoRa driver and HMAC in plain C on STM32.
- **Bugs we hunted.** Spurious LoRa re-arm calls. SPI bus contention between ISR and tasks. Keil toolchain friction.
- **What worked.** The state machine survived every test *and* the entire platform migration — unchanged.
- **Lesson.** Diagnostic counters > guesswork. Build for observability from day one.

M3 · SIMULATION

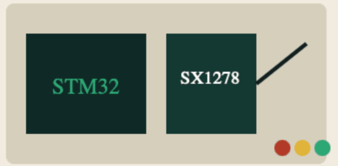
Wokwi ESP32

browser-based circuit sim

→

FINAL · HARDWARE

STM32 + SX1278



breadboard prototype

Three people, three pillars.

Ownership areas, not silos — everyone understood the whole system.

ARCHITECTURE

Ziad Gaballah

Real-time OS architecture & the traffic-light state machine.

HARDWARE

Khadeejah Iraky

Hardware integration & the debugging that came with it.

PROTOCOL & DOCS

Hamza Kamel

Wireless protocol design & documentation.

Clear next steps, in priority order.

01	**Fix HMAC verification** Code is present — the verify logic needs one correction. Not a redesign.	HIGHEST
02	**RSSI proximity detection** Use signal-strength trend to know the ambulance is approaching vs. leaving.	HIGH
03	**Battery power + range test** Validate operation at > 100 m line-of-sight.	MEDIUM
04	**Solder real antennas** Replace the through-hole wire antenna with a tuned soldered one.	MEDIUM
05	**Chain a second intersection** Demonstrate a true corridor "green wave" — the real product story.	STRETCH

A useful subset of a real system, for $30 and a few weeks.

- The core demo **works authentically**: press → clearance → response → failsafe.
- The **state-machine and safety design** survived a full platform migration.
- We scoped **honestly**; we know **exactly** what's left.

"GreenWave proves that a useful subset of an emergency-vehicle preemption system can be built from $30 of parts and a few weeks of student work."

Thank you — questions?

Hamza Kamel · Khadeejah Iraky · Ziad Gaballah